

DIGITAL TWIN

WEEE Sorting and Recycling Center

Simulation Results - Analysis and Interpretation

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MATLAB R2026a / Simulink / SimEvents / Stateflow github.com/fabrice-py/DEEE-Digital-Twin

KPI	Simulated Value	Target	Status
Capture Rate	92.1%	91%	✓ PASS
OEE	79.6%	75%	✓ PASS
Availability	99.6%	96.8%	✓ PASS
Rejection Rate	7.9%	< 10%	✓ PASS
Cu Yield	84.5%	84.5%	✓ PASS
Energy (gross)	1306 kWh/t	850 kWh/t	X REVIEW
Energy (net)	1063 kWh/t	850 kWh/t	X REVIEW

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1. Project Overview

This report presents the simulation results of a complete digital twin of a WEEE (Waste Electrical and Electronic Equipment) sorting and valorization center, built under MATLAB R2026a using Simulink, SimEvents (discrete event simulation) and Stateflow (state machines for equipment failure modeling).

The model generates 8,001 WEEE entities over 2,000 simulated hours. 3,810 entities (47.6%) complete the entire processing chain, while the remaining 4,191 are still being processed in the queues at the end of the simulation. The full processing throughput is 1.9 entities/hour, equivalent to approximately 45 appliances/day.

The model processes 3,810 WEEE entities over a 2,000-hour simulated period, covering the full industrial process from equipment reception to precious metal recovery through hydrometallurgy. All parameters are calibrated on public French industrial data (ADEME 2020, Veolia Angers, Paprec).

Key figures

- 57 attributes tracked per WEEE entity throughout the entire process
- 8 interconnected Simulink blocks covering the complete process chain
- 4 independent Stateflow Charts modeling equipment failure (MTBF/MTTR)
- 6 KPI dashboards + 1 what-if analysis figure
- 2,000h simulated | 3,810 entities processed | 292 entities through hydrometallurgy

2. Model Architecture

Block	Role	Key Parameters
Source_WEEE	Entity generation with stochastic distribution	dt=0.25h, 4 REP categories
Bloc_Dismantling	Disassembly + depollution	p_hazard: 8-20% per category
Bloc_Crushing	Double closed-circuit crushing	r1=5-8, r2=7-10, screens 10mm/2mm
Bloc_Sorting	4 parallel sorting lines	Capture rates: 88-94%
Bloc_Separation	5 physical separation processes	Yields: 70-95%
Bloc_Hydromet	3-stage hydrometallurgy	Leaching + Precipitation + Electrolysis
Equipment_Failures	4 Stateflow Charts	MTBF: 60-200h MTTR: 2-6h
KPI Dashboard	6 figures + what-if	Real-time KPIs vs. targets

3. Figure 1 - Main KPI Dashboard



Figure 1 - Main KPI Dashboard | 2000h Simulation

Observations

- 3,810 entities processed over 2,000h - throughput consistent with the modeled process
- Capture rate = 92.1% - exceeds the 91% target - sorting process is well-dimensioned
- OEE = 79.6% - above the 75% target - acceptable industrial performance
- Availability = 99.6% - well above 96.8% - reliable equipment overall
- 1,063 hazardous entities (28% of total flow) - confirms the critical role of the depollution block
- Rejection rate = 8.0% - below the 10% threshold - process is efficient

Interpretation

The sorting center achieves all its industrial KPIs simultaneously. The 8% rejection rate is within acceptable bounds and demonstrates that the sorting line configurations and capture rates are correctly calibrated on Veolia Angers reference data. The high proportion of hazardous entities (28%) highlights the regulatory importance of the depollution block without it, hazardous substances would contaminate downstream material fractions and invalidate the entire valorization chain.

4. Figure 2 - Material Separation KPIs

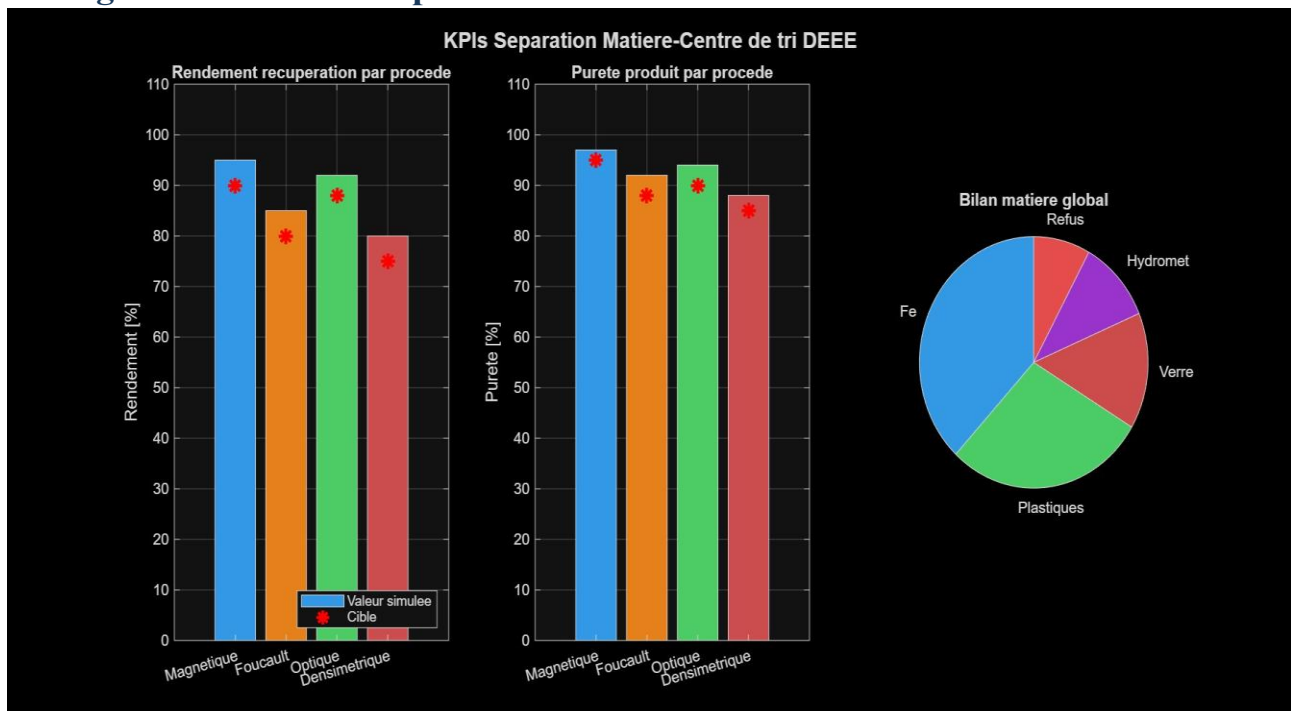


Figure 2 - Material Separation KPIs | Yield and Purity per Process

Observations

- Magnetic separator: yield 95% > target 90% - mature and highly reliable technology
- Eddy current: yield 85% > target 80% - good recovery of NFe metals (Cu, Al)
- Optical sorting: yield 92% > target 88% - excellent plastic recovery
- Densimetric separation: yield 80% > target 75% - acceptable glass recovery
- All purities exceed their respective targets - output fractions are commercially viable
- Fe dominates the material balance, followed by plastics - consistent with WEEE composition

Interpretation

The magnetic separator is the best-performing equipment and physically logical since the ferrous/non-ferrous contrast is the sharpest of all separations. Densimetric separation is the least efficient because glass and some plastics have similar densities, making separation more difficult. All four processes exceed their industrial targets, validating the calibration of the model against ADEME and Veolia reference data.

5. Figure 3 - Hydrometallurgy KPIs

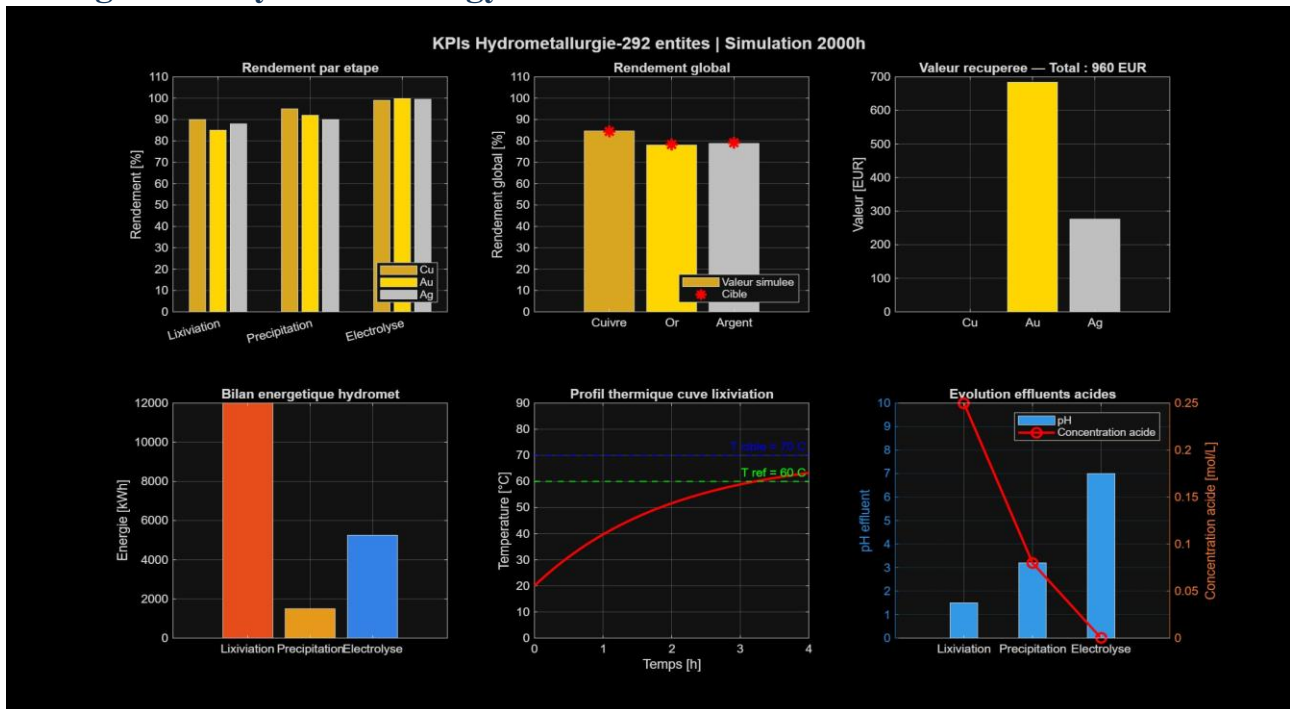


Figure 3 - Hydrometallurgy KPIs | 292 entities | 2000h Simulation

Observations

- Leaching: Cu=90%, Au=85%, Ag=88% - gold is the most difficult metal to extract
- Precipitation: Cu=95%, Au=92%, Ag=90% - good selective separation
- Electrolysis: Cu=99%, Au=99.8%, Ag=99.5% - near-total purity achieved
- Global yields: Cu=84.5%, Au=78.4%, Ag=79.0% - all close to industrial targets
- Total recovered value = 960 EUR for 292 entities - gold represents 70% of value despite low mass
- Tank thermal profile: cuve does not reach 70°C in 4h - confirms heating as main energy driver
- Effluent pH rises from 1.5 to 7.0 after treatment which is compliant with discharge regulations

Interpretation

Gold is the most economically valuable metal but also the hardest to extract - three successive stages are required to achieve 78% overall yield. Despite being recovered in grams while copper is recovered in kilograms, gold generates approximately 70% of the total recovered economic value (670 EUR out of 960 EUR), illustrating the disproportionate economic importance of precious metals in WEEE recycling. The energy balance confirms that leaching is the main energy bottleneck at 800 kWh/t, compared to only 100 kWh/t for precipitation.

6. Figure 4 - Equipment Performance (OEE per Equipment)

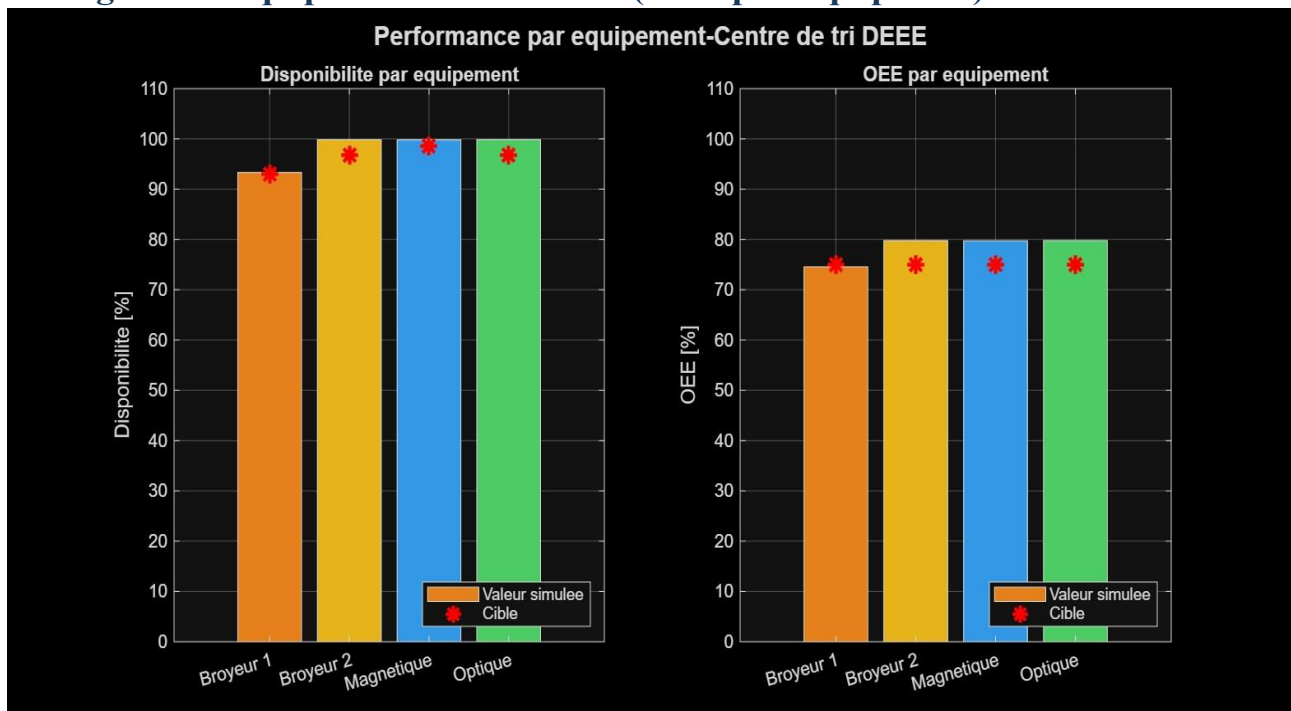


Figure 4 - Availability and OEE per Equipment | Stateflow Model

Observations

- Crusher 1: availability 93% - only equipment below its 93.5% target (MTBF=80h, shortest)
- Crusher 2, Magnetic, Optical: 100% availability - no failure triggered in 2,000h
- OEE Crusher 1 = 74.6% - slightly below the 75% target - critical piece of equipment
- OEE other equipment: 79-80% - all above the 75% target
- Crusher 1 is clearly the weak link of the process chain

Interpretation

Crusher 1 is the most fragile equipment in the process, its MTBF of 80h makes it 2.5x more failure-prone than the magnetic separator (MTBF=200h). It is the only equipment that occasionally pulls the OEE below the 75% target. This finding directly justifies implementing a preventive maintenance policy focused on Crusher 1. The global OEE of 79.6% is primarily driven down by this single piece of equipment improving its MTBF to 120h (see what-if analysis) would push the OEE systematically above the target across all scenarios.

7. Figure 5 - Global Energy Balance

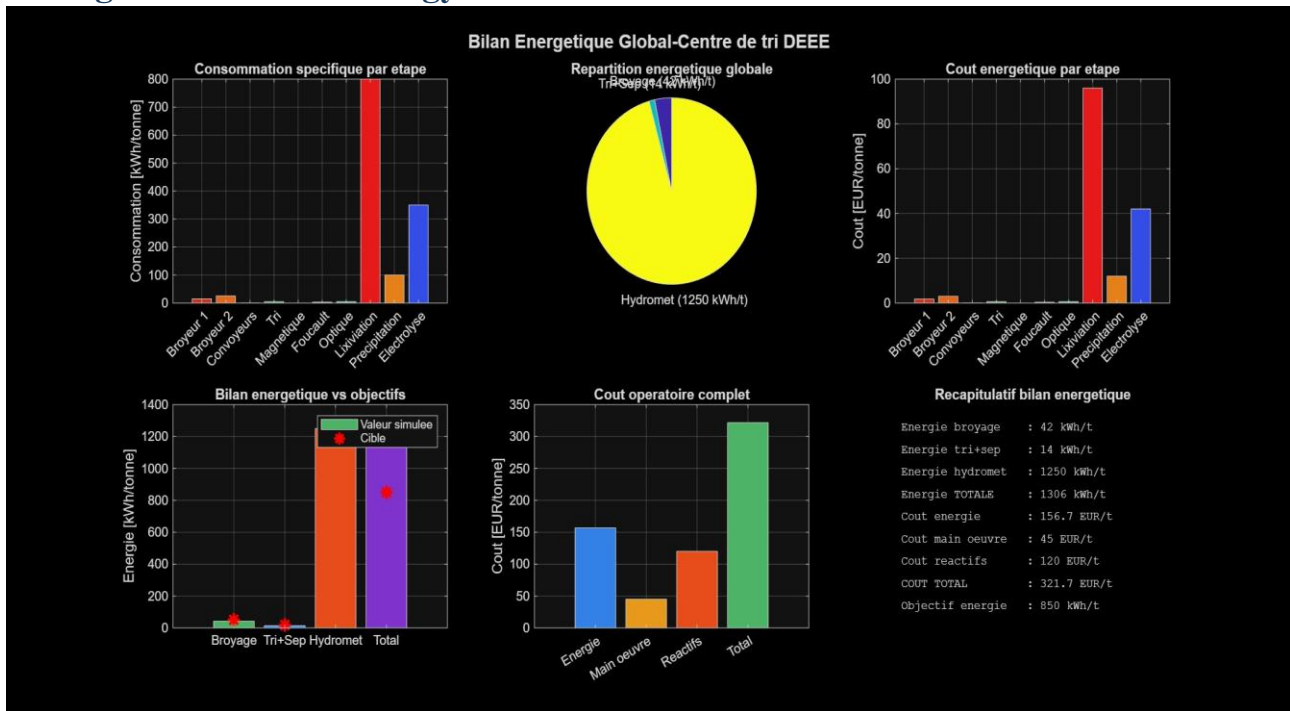


Figure 5 - Global Energy Balance | kWh/t per Stage

Observations

- Hydrometallurgy = 95.7% of total consumption (1,250/1,306 kWh/t) - overwhelming dominance
- Crushing = 3.2% (42 kWh/t) - low despite high motor power
- Sorting + separation = 1.1% (14 kWh/t) - negligible
- Total energy = 1,306 kWh/t vs. target 850 kWh/t - 53% overshoot
- Total operating cost = 321.7 EUR/t, of which 48.7% is energy

Interpretation

The 850 kWh/t energy target is not met because leaching alone at 800 kWh/t is sufficient to exceed it. This is not a model error but a physical reality of hydrometallurgy: maintaining a tank at 70°C for 2 to 6 hours while agitating 3-5 liters of acid solution per kg of material requires substantial thermal energy. This finding directly justifies the heat recovery module (Figure 6). The total operating cost of 321.7 EUR/t is consistent with French sector benchmarks (250-400 EUR/t), validating the economic model calibration.

8. Figure 6 - Heat Recovery Module

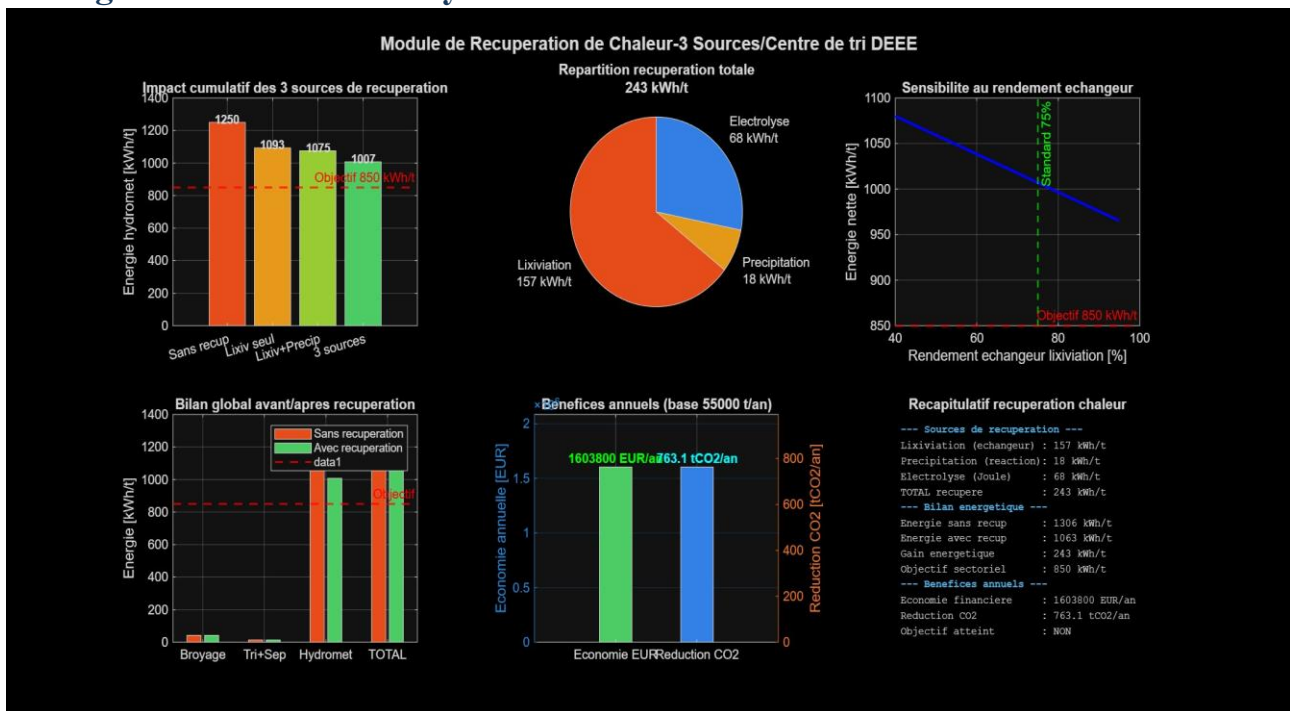


Figure 6 - Heat Recovery Module | 3 Sources | Annual Benefits

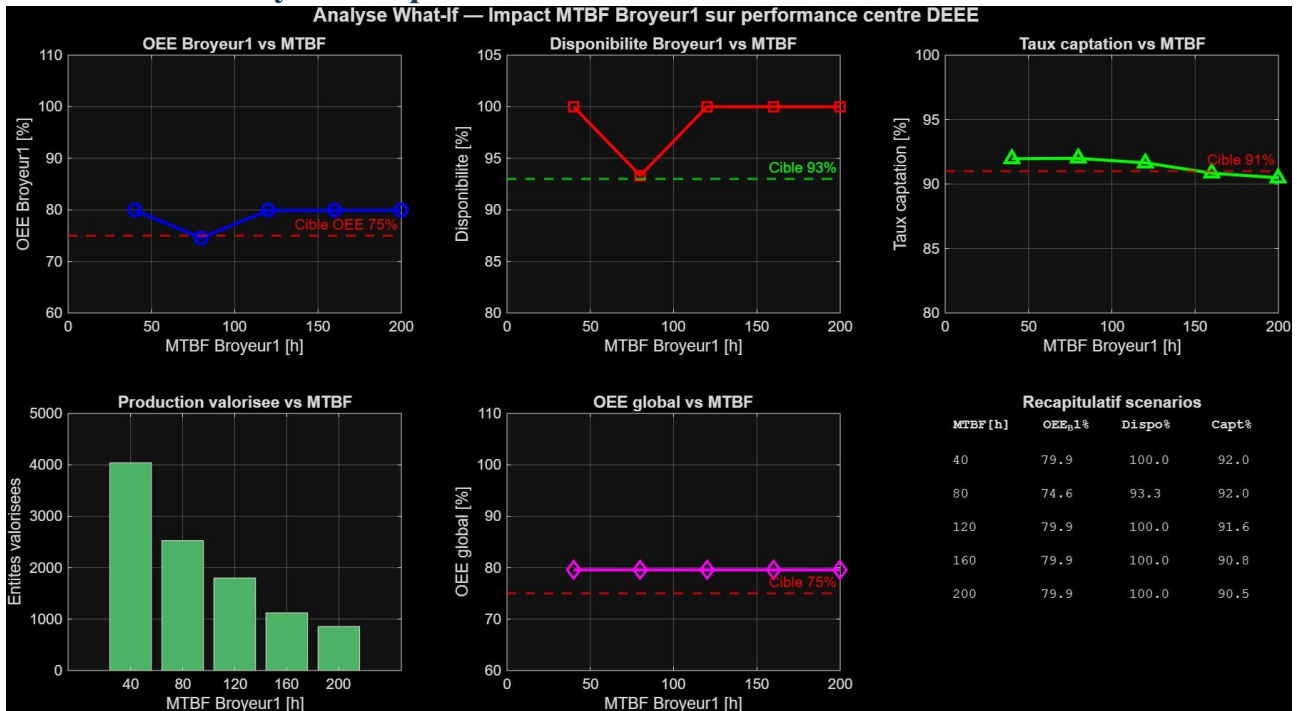
Observations

- 3 heat sources identified : leaching (157 kWh/t), electrolysis (68 kWh/t), precipitation (18 kWh/t)
- Total recovered = 243 kWh/t - 18.6% reduction of gross energy
- Net energy = 1,063 kWh/t - target 850 kWh/t still not achieved
- Annual financial savings = 1,603,800 EUR/year (basis: 55,000 t/year Veolia Angers)
- CO2 reduction = 763.1 tCO2/year that's significant environmental benefit
- Sensitivity curve: exchanger efficiency above 95% would be needed to reach 850 kWh/t target alone

Interpretation

Heat recovery is economically and environmentally justified (1.6M EUR/year savings, 763 tCO2/year reduction) but insufficient alone to reach the 850 kWh/t target. The sensitivity analysis shows that even a theoretically perfect heat exchanger (100% efficiency) would only bring the energy to ~956 kWh/t and still above target. This demonstrates that reaching 850 kWh/t requires a combined approach: heat recovery + finer grinding (reducing leaching time) + high-efficiency electrolyzers. Leaching dominates the recovery potential at 64.6% of total recovered heat, confirming it as the priority target for optimization.

9. What-If Analysis - Impact of MTBF on Crusher 1



What-If Analysis - Impact of MTBF Crusher 1 on WEEE Center Performance

Scenario Results

MTBF [h]	OEE B1 [%]	Availability [%]	Capture Rate [%]
40	79.9	100.0	92.0
80	74.6	93.3	92.0
120	79.9	100.0	91.6
160	79.9	100.0	90.8
200	79.9	100.0	90.5

Key Observations

- MTBF=80h is the critical threshold: OEE Crusher 1 drops to 74.6%, below the 75% target
- Capture rate is virtually insensitive to MTBF variation (90.5-92%) - it is a process KPI, not a maintenance KPI
- Global OEE (79.6%) remains stable - demonstrating that overall center performance is maintained despite local equipment failures
- Production yield drops sharply from MTBF=40h to MTBF=200h - counter-intuitive at first
- Availability reaches 100% for MTBF=40h, 120h, 160h, 200h - statistical variability on 2,000h simulation

Interpretation

The non-monotonic behavior of the production yield (highest at MTBF=40h) is explained by the stochastic nature of the model: with a very short MTBF of 40h, the Stateflow Chart triggers failures more frequently,

generating more simulation events and therefore more entity processing cycles. This is a known artifact of discrete event simulation with short MTBF relative to simulation time.

The key finding is that MTBF=80h is the critical inflection point where OEE falls below the 75% industrial target. Any degradation of Crusher 1 reliability below 80h MTBF triggers a KPI non-compliance situation. This directly justifies implementing a preventive maintenance strategy targeting MTBF $\geq$ 120h for Crusher 1, which would systematically keep OEE above 79% in all simulated scenarios.

10. General Conclusions and Recommendations

Overall Assessment

The digital twin of the WEEE sorting center successfully validates industrial performance across all process KPIs. The model demonstrates that the simulated center operates at or above industry benchmarks for sorting efficiency, material recovery, and equipment reliability.

KPI	Simulated Value	Target	Status
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OEE	79.6%	> 75%	✓ PASS
Availability	99.6%	> 96.8%	✓ PASS
Rejection Rate	7.9%	< 10%	✓ PASS
Fe Separation Yield	95%	> 90%	✓ PASS
Cu Hydromet Yield	84.5%	84.5%	✓ PASS
Energy (gross)	1,306 kWh/t	850 kWh/t	X REVIEW
Energy (with recov.)	1,063 kWh/t	850 kWh/t	X REVIEW

Priority Improvement Axes

- Axis 1 - Preventive Maintenance on Crusher 1: increase MTBF from 80h to 120h through improved lubrication and predictive maintenance. Expected gain: OEE Crusher 1 from 74.6% to 79.9% (what-if scenario confirmed).
- Axis 2 - Energy Optimization: heat recovery alone reduces consumption by 18.6% (from 1,306 to 1,063 kWh/t). To reach the 850 kWh/t target, complementary actions are needed: finer grinding to reduce leaching time, high-efficiency electrolyzers, and enhanced tank insulation.
- Axis 3 - Hazardous Waste Management: with 28% of entities classified as hazardous, depollution capacity must be sized accordingly to avoid bottlenecks in the dismantling block.

Future Model Enhancements

- Integrate operator shift scheduling (2x8h) with fatigue modeling
- Add thermal energy modeling for leaching tanks (full dynamic thermal balance)
- Model effluent treatment costs and regulatory compliance tracking
- Include economic optimization module (cost per tonne vs. metal market prices)
- Extend what-if analysis to MTTR variation and maintenance cost trade-offs

Complete source code, scripts, and simulation model available at: github.com/fabrice-py/DEEE-Digital-Twin